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Studierichting: Informatica
Oefeningenreeks 4A nr. 44.4

$$\int \frac{x^4+x^3-3x+5}{x^3+2x^2+2x+1}dx$$

$$x^3+2x^2+2x+1=(x+1)\left(x^2+x+1\right)$$

$$\frac{x^4+x^3-3x+5}{x^3+2x^2+2x+1}=x-1+\frac{6-2x}{x^3+2x^2+2x+1}$$

$$\frac{6-2x}{(x+1)\left(x^2+x+1\right)}=\frac{A}{x+1}+\frac{Bx+C}{x^2+x+1}$$

$$A=\left.\frac{6-2x}{x^2+x+1}\right|_{x=-1}=8$$

$$Bx+C=\left.\frac{6-2x}{x+1}\right|_{x=-\frac{1}{2}+i\frac{\sqrt{3}}{2}}$$

$$=\frac{7-i\sqrt{3}}{\frac{1}{2}+i\frac{\sqrt{3}}{2}}$$

$$=\frac{2\left(7-i\sqrt{3}\right)}{1+i\sqrt{3}}$$

$$=\frac{2\left(7-i\sqrt{3}\right)\left(1-i\sqrt{3}\right)}{\left(1+i\sqrt{3}\right)\left(1-i\sqrt{3}\right)}$$

$$=\frac{2\left(4-8i\sqrt{3}\right)}{4}$$

$$=2-4i\sqrt{3}$$

$$\Rightarrow -\frac{B}{2}+C+i\frac{B\sqrt{3}}{2}=2-4i\sqrt{3}$$

$$\Rightarrow \frac{B\sqrt{3}}{2}=-4\sqrt{3}\Rightarrow B=-8$$

$$\Rightarrow -\frac{-8}{2}+C=2\Rightarrow C=-2$$

$$\Rightarrow \int\left(x-1+\frac{8}{x+1}+\frac{-8x-2}{x^2+x+1}\right)dx$$

$$=\int(x-1)\,dx+8\int\frac{dx}{x+1}-\int\frac{8x+2}{x^2+x+1}dx$$

$$8x+2=4\,(2x+1)-2$$

$$=\frac{x^2}{2}-x+8\ln|x+1|-4\int\frac{2x+1}{x^2+x+1}dx+2\int\frac{dx}{x^2+x+1}$$

$$=\frac{x^2}{2}-x+8\ln|x+1|-4\ln\left(x^2+x+1\right)+2\int\frac{dx}{\left(x+\frac{1}{2}\right)^2+\frac{3}{4}}$$

$$= \frac{x^2}{2} - x + 8\ln|x+1| - 4\ln\left(x^2+x+1\right) + \frac{4}{\sqrt{3}}\,\mathrm{Bgtan}\left(\frac{2x+1}{\sqrt{3}}\right) + k$$

References